logic so as to allow testing, even if not all of them are as tightly integrated with testing framework as Rails.

For example, the JavaScript code you wrote in Chapter 9, *Testing JavaScript: Integration Testing*, on page 181, uses the DOM as the source of truth for the order of the tasks. The code queries the DOM to find the previous and next tasks when the time comes to reorder them. While that might be fine for something as simple as reordering tasks, it's already a little tricky to read if you don't know the expected DOM structure. Adding simple extensions—like hiding the up or down buttons for the first and last tasks—isn't clear. In this chapter you'll take another swing at that code, attempting to drive it from unit testing rather than integration testing.

Setting Up JavaScript Unit Tests

This is going to be a bit involved, I'm afraid. Rails 5.1 uses the Webpacker gem, which allows you to bypass the Rails asset pipeline and use Webpack—an asset-bundling tool that is in common use by JavaScript developers. Webpack allows you to do all the same things the old Rails asset pipeline does: module lookup, compilation, conversion to a single minified file, and the like. But Webpack is more integrated with current JavaScript tools and makes it much easier to integrate with JavaScript code that is distributed using the Node Package Manager (NPM).[1]

That's the good news. The less-good news is that Webpack configuration can be really complicated, especially if you're a Rails developer and are used to convention over configuration. The Rails team has put a lot of effort into Webpacker to control the complexity of Webpack configuration, but you still sometimes need to dig into the details.

Also note that Webpacker is in version 3.2.1 as I write this. It has had two or three major revisions since I started working on this book, each of which substantially changed the configuration files. It's highly likely that there's been another change or two between when I wrote this and when you're reading it.

In this chapter I'll use the Karma test runner,[2] which runs from the command line but executes the tests against browsers and plays reasonably nicely with

1. https://www.npmjs.com
2. https://karma-runner.github.io